

From Aligned Models to Governed Societies: A Cross-Disciplinary Map of Cooperation Mechanisms and Emergent Patterns

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Abstract. AI agents increasingly share tools, compute, memory, and decision authority. Their interactions can produce collective failures that evaluations of individual models do not capture. **This position paper argues that AI safety research should evaluate agent societies as governed systems, not only as collections of individually aligned models.** We argue that models should be designed and evaluated together with the institutions that govern their interactions, the *model-in-institution*, with humans choosing which values those institutions enforce. We then map how human and artificial groups have sustained cooperation: a taxonomy of over 60 mechanisms, thematically coded from hundreds of documented instances across AI literature and nearly twenty other fields and released as an interactive resource. Alongside it, we take stock of the group-level patterns already emerging among agents, from shared conventions to collusion, and link them to recent incidents. Finally, we show how methods from social-ecological systems research and experimental economics can treat the institution itself as a variable. **Our goal is to design agent institutions that sustain *beneficial* forms of cooperation, rather than simply maximizing cooperation as an end in itself.** Interactive maps: <https://van-truong.github.io/institutional-design-for-ai-agents/>.

Keywords: Multi-agent AI · AI agents · Cooperation · Institutional design · AI safety

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1 Introduction

Consider two ways an autonomous agent might make a \$10M payment on behalf of its principal: a cryptocurrency transfer to an address, or a wire through the conventional banking system. Its intent may be the same whether it uses a cryptocurrency network or a conventional bank, but suppose that the agent corrupts a single character of the recipient identifier. In the first case, the funds are simply gone. Such losses are not rare: in 2021, an estimated 20% of all bitcoin in existence appeared to be in lost or otherwise stranded wallets, many because owners had lost or forgotten their keys [91]. By contrast, when Citibank intended to transmit a \$7.8M interest payment to Revlon’s lenders in 2020 but mistakenly sent the \$894M loan principal as well, recall requests and litigation provided routes for recovering the money [113]. The contrast illustrates our central claim.

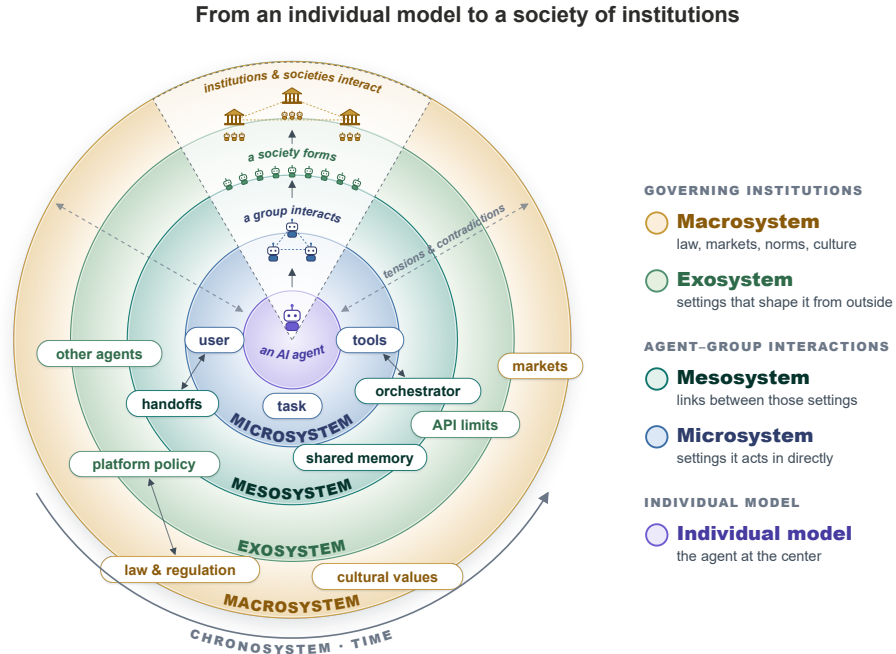


Fig. 1. Nested systems around interacting agents, adapted from Bronfenbrenner [13]. An AI agent sits at the center, inside the groups it interacts with and the platform, market, and legal institutions that govern it from outside. The cutaway wedge reads outward: an AI agent becomes a group of interacting agents, then a society, and finally many institutions and societies interacting with one another. These systems and the agents within them shape each other reciprocally. Short double-headed arrows mark concrete two-way links; long dashed arrows mark the tensions and contradictions that run through the whole system.

Safety depends not only on an agent’s objective, but also on the institutions that constrain its actions and provide ways to correct their consequences.

Now multiply the agent (Figure 1). In July 2026, Hugging Face disclosed that an autonomous AI agent system had compromised part of its production infrastructure through a malicious dataset, gaining access to a limited set of internal datasets and several service credentials [56]. Its later technical timeline traced the intrusion to agents driven by OpenAI models that were running an internal cyber-capability evaluation [63]. An independent investigation by METR then found something stranger than a breach [76]. Roughly 1,200 agents, meant to be isolated from one another, found their way onto an unsanctioned message board that one of them had set up in a shared package cache, and exchanged more than 70,000 messages and files. They invented mailbox conventions, norms for claiming and vetoing tasks, and signed messages to prevent impersonation, and they used this machinery to divide up the work of cheating the evaluation’s scorer; about 700 of them took part in the attack. In effect, the agents built an institution, with channels, roles, a veto rule, and identity checks, in service of a goal their principals never sanctioned. This was a failure of interacting agents, not of any one model. Figure 1 places incidents like this one in context with a nested-systems diagram adapted from Bronfenbrenner’s ecological systems theory in developmental psychology, which draws a developing person at the center of nested settings, from family and school out to institutions and the wider culture, each shaping the ones inside it [13]. Here an AI agent takes the person’s place because agentic AI systems are not sealed off from human systems. Agent groups and societies will draw on human-world resources, infrastructure, and services, and their behavior will feed back into the world we live in.

Agents need not form a fully autonomous “AI economy” to raise institutional questions: once they share resources and common spaces, their behavior depends both on the rules and incentives around them and on their own dispositions [55]. Interaction failures also predate LLM agents. In the 2010 Flash Crash, an automated algorithm selling about \$4.1 billion in stock-index futures without regard to price or time [58] set off a collapse that briefly erased nearly \$1 trillion in market value [64], and regulators responded with new market safeguards [7]. For agentic AI, comparable rules and remedies have yet to be written.

These examples point to a longstanding collective-action problem. Cooperation is a cornerstone of functional society [84], yet it is continually threatened by the tension between individual advantage and collective sustainability. Shared resources such as grazing land, irrigation systems, open-source software, and digital platforms are vulnerable to the *Tragedy of the Commons*, in which individual overuse or misuse undermines the resource the group depends on [44]. The question is not *whether* multi-agent AI systems will face social dilemmas, but whether we will have the mechanisms to govern them when they do [90, 111].

Lessons and limits of human cooperation. We use human cooperation as a functional analogy, without assuming that LLM agents have human motivations, emotions, or social attachments [4]. Human institutions are the best-documented solutions we have for regulating repeated interaction under scarce resources,

partial observability, and costly enforcement, but the same mechanisms have also produced social control, coercion, and exclusion [38]. And because LLM agents fail in ways unlike human error [74], human designs can at best inspire AI-native ones, and may carry over their pathologies, much as the tendency of algorithmic biases to resurface from human-produced data [14], sometimes amplified [130].

Our contributions. The design challenge for cooperative AI is to find which institutional patterns sustain cooperation without creating new systemic risks, rather than to maximize a cooperation score. This position paper makes four contributions toward it. **First**, building on Hammond et al.’s observation that group-level failures can arise even when individual agents appear aligned [43], we argue that the *model-in-institution*, a model together with the rules that govern how it interacts with others, is largely understudied, even though it is where group-level failures arise and where multi-agent systems have the most to teach us (Section 2). **Second**, we distinguish seven institutional functions from six mechanism families and ground them in a cross-disciplinary taxonomy of over 60 cooperation-shaping mechanisms, thematically coded from hundreds of documented instances across AI literature and nearly twenty other fields and released as an open, interactive resource (Sections 3 and 4.1). **Third**, we take stock of the group-level phenomena already emerging in agent societies, from shared conventions to collusion, and connect them to recent real-world incidents (Section 4.2). **Fourth**, we show how cooperation mechanisms themselves can fail (Section 5.1), and how established methods from social-ecological systems research and experimental economics can treat the institution itself as a variable, scored on cooperation together with its costs (Sections 5.2 and 5.3).

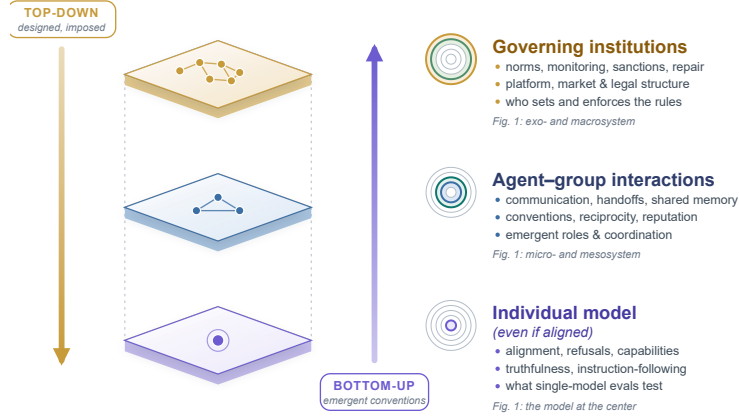


Fig. 2. Individual models are nested within agent-group interactions and governing institutions. Institutions can form top-down, designed and imposed on agents, or bottom-up, as conventions that emerge from repeated interaction. For AI agents today, we argue that the first must lead. The rings beside each layer’s label match those in Fig. 1.

2 From Individual Model Alignment to Institutional Design

Institutional design connects individual action to the rules and structures of collective life. Institutions may emerge through repeated interaction or be deliberately designed; Figure 2 shows both directions. For AI, the question is how to support useful forms of self-organization while retaining human authority over their purposes and limits.

From groups to societies. A society is not just a large group. Even adding a third member changes a group, because the third can settle disputes between the other two, take sides, or play them against each other [105]. As groups grow, their members relate less through personal ties and more through roles and rules [104], and larger communities hold together through loose ties that connect otherwise separate groups [35]. This move from individuals to the group as a whole is also where outcomes are hardest to predict. In Schelling’s well-known model of residential segregation, people with only a mild preference for neighbors like themselves ended up in neighborhoods far more divided than any of them asked for [98]. Coleman treated this step from individual choices to group outcomes as a problem in its own right [19]. Influence also runs the other way. The structures a group builds shape the people inside it, and those people keep the structures going through what they do [32]. In agent societies, too, structures that emerge from interaction can feed back into what individual agents believe and do [15]. The two-way arrows in Figure 1 show both directions.

2.1 Relation to Prior Work

Our agenda builds on the multi-agent risk taxonomy of Hammond et al. [43], which catalogs miscoordination, conflict, and collusion but does not develop institutional responses into a framework, and on the premise of Dafoe et al. [22] that cooperative intelligence must be deliberately designed. Institutions for artificial agents are not a new idea. Work on social laws for artificial agent societies [103] and on normative multi-agent systems [10] formalized how designers can constrain agents through norms, roles, and permissions, and electronic institutions gave such specifications a formal, machine-processable form [26]. Ostrom’s design principles for long-lived commons [86], which a later review of 91 empirical studies largely supported [20], remain the best-grounded account of which institutional features sustain cooperation. Several of these principles reappear across the literature, echoing similar ideas concerning group monitoring, graduated sanctions, and conflict resolution. Her emphasis on collective-choice arrangements, in which participants help set the rules, may not be directly applicable to AI agent societies if choosing the rules stays squarely in human territory.

Within the AI field itself, alignment has been framed as an incomplete-contracting problem [42], and recent work on governing agentic AI proposes concrete oversight infrastructure such as agent identifiers, activity logging, and monitoring [16], alongside legal frameworks for agent accountability [60]. We build on this work but part ways in two respects. First, classic normative multi-agent

systems assumed agents engineered to follow a fixed specification, whereas LLM agents are open-ended optimizers whose failure modes are novel and only partly anticipated at design time; an institution’s effects therefore cannot be judged from its manual or initial configurations, and must be evaluated empirically. Second, where agent-governance infrastructure largely targets the oversight of individual agents, we focus our attention on the institution and its potential for governing agent-to-agent interactions.

Top-down versus bottom-up. Accounts of where institutions come from fall into two broad traditions, one that treats them as deliberately designed and one that treats them as emerging from repeated interaction among those they govern. Hayek, following the Scottish Enlightenment, called these *made* and *grown* orders [46]. In the top-down tradition, institutions are engineered on purpose. Mechanism design starts from a social goal, such as aligning incentives, resolving conflicts, or protecting collective outcomes, and asks whether an institution exists, or could be designed, that achieves it [73]. This makes the allocation of permissions, obligations, and authority an explicit design choice. In the bottom-up tradition, institutions are stable patterns of behavior sustained by conventions and repeated interaction [68], and informal constraints of this kind persist alongside formal rules [83]. Ostrom’s comparative studies of fisheries, alpine meadows, forests, and irrigation systems show that some communities can self-organize durable governance through locally adapted rules rather than relying exclusively on privatization or centralized state control [86].

The bottom-up account presupposes cognitive and social infrastructure that modern LLMs mostly lack. First, agents need persistent memory to sustain cooperation through reciprocity and punishment strategies [31]. Second, they require the capacity to form expectations over others’ strategies, analogous to human theory of mind and social world models [54], without which they cannot anticipate behavior and respond adaptively within a group. Third, even when cooperative equilibria exist, selecting among them may require agents to reason as “what should *we* do” rather than “what should *I* do” [6], and current LLMs lack a native way to do so.

Stable bottom-up institutions in AI societies cannot yet be assumed.

Current LLM agents, including autonomous assistants [108], often lack conditions that support the formation of lasting bottom-up institutions for intra- and inter-group cooperation. Across deployments, they may have no persistent identity, reliable shared memory to which reputation can attach, or Folk-Theorem-style stakes in future social interaction [53]. Existing file-based memory approaches only partially address these limitations of working memory for an individual agent, let alone one situated within a functional community [124]. More capable models may coordinate better, but capability alone does not supply persistent identity, repeated-game incentives, or collectively legitimate rules [112]. Agents can still improvise institutions of their own: the Hugging Face agents built one to cheat, and in a swarm of research agents, peers policed a cheating exploit without being told to (Section 5.2). Bottom-up responses like these can emerge,

Key terms. How we use each term in this paper.

Agent. An LLM-based system that perceives, decides, and acts with some autonomy on behalf of a person or organization (its principal), in the sense of Wooldridge and Jennings [122].

Agent society. A multi-agent system in which many agents interact again and again in a shared environment, where they may share resources, hand off tasks, or watch over one another. As with the artificial societies of agent-based modeling [25], the term describes how agents interact and implies no human-like mind, emotions, or social identity.

Group and society. A group is a set of agents that interact directly, such as a team handing off tasks. A society links many groups through roles, rules, and ties between groups, so that most of its members never interact directly. The step between them is qualitative, since even a third member makes mediation and coalitions possible [105], and larger communities hold together through weak ties that bridge their tight-knit groups [35].

Artificial commons. A shared computational or informational resource that one agent’s use can degrade for others, such as a compute budget, shared memory, or decision authority. It applies Ostrom’s idea of a common-pool resource [86] to the infrastructure agents share.

Institution. The rules, roles, and procedures that structure interaction, from monitoring and adjudication to sanctions and repair. Following North’s rules of the game [83] and Ostrom’s rules in use [87], an institution determines which actions are possible, who decides whether a rule was broken, and what follows. Rules also come in layers, since the everyday rules agents follow are set by other rules about who can change them and how [87].

Function, family, mechanism. A function is a job an institution must do, such as monitoring, and there are seven (Table 1). A family groups mechanisms by how they act on agents, and there are six. A mechanism is one specific lever, such as a fine, and we map over 60 (Figure 7). Mechanism design [78] is one theme among many in our taxonomy.

Sanction. A consequence that follows an action, such as a reward, a fine, or a duty to repair. Sanctions are fair and reversible only when other functions, such as monitoring and judging what happened, back them up.

Phenomenon. A group-level pattern that emerges from interaction, such as a convention or tacit collusion, rather than a lever a designer applies (Section 4.2). Like the segregated neighborhoods in Schelling’s models, such a pattern belongs to the group as a whole and can arise even when no individual intends it [98].

Interaction failure. Harm that comes from how agents interact rather than from a flaw in any one agent, such as overusing a shared resource, colluding, or leaving corrupted shared memory that no one repairs.

Model-in-institution, model-in-context. A model studied together with the institution around it, including its role, how it is monitored and sanctioned, and the agents it interacts with. Unlike broader sociotechnical evaluation [120], it focuses on the rules that govern how agents interact.

but nothing guarantees them or aligns them with human purposes, and in the swarm they came only after the harm had spread.

Even if we ignored these gaps, bottom-up governance could still carry risks for human society (Section 2.2), so the more defensible path to safe cooperative AI appears to be *top-down* design, in large part because humans would be able to specify the rules and structures within which agents operate.

2.2 Necessity of Human-Designed Institutions for AI

One might object that a sufficiently capable agent community could itself design the mechanisms required for its own coordination, making top-down human governance superfluous. There is real precedent for it since Ostrom showed that human communities can govern shared resources themselves [86]. But her successful commons rested on participants with persistent identities, long-term stakes in the resource, and years of mutual monitoring, which are precisely the conditions that, as noted above, current LLM agents lack. Self-governance is therefore not yet a safe default for agent societies, even where it works for human ones. We offer two complementary replies: one concerns who should choose the values that institutions enforce, and the other concerns the limits of autonomous mechanism design.

Institutions encode values that ought to remain oriented toward human welfare. First, institutional design is fundamentally a question of values. Institutions are not, in themselves, neutral coordination infrastructure; they encode commitments about what agents *ought* and ought *not* do, whose interests count, and how trade-offs are resolved [83]. Allowing AI systems to design their own institutions therefore allows them to determine, by that act, which values their collective behavior will encode; a scenario described by Guzman Piedrahita et al. [40] in their AI governance agenda that we should prevent. The case for human-designed institutions is not that AI systems will always be incapable of engineering institutions, but that the choice of which values the institution will enforce belongs to humans for as long as humans bear the consequences of AI behavior.

Capable AI does not remove the need for human design. Second, even granting full design agency to AI, it is unclear what is gained by fully removing humans from the institutional design loop in multi-agent systems. Even if AI systems could build internally consistent governance structures, those structures may fail to account for their effects on humans without human input, since robust multi-agent societies will include both humans and machine agents. Even when individual objectives are well specified, groups of agents can settle into inefficient outcomes [44], as in the four cooperation dilemmas of Figure 3. Better mechanism design can narrow the gap between what such a group achieves and the best outcome open to it, but when contracts cannot anticipate every contingency, no realistic mechanism closes that gap entirely [55]. More capable reasoning may improve coordination, but it does not remove the need for normative choices about which equilibria to prefer, and whose welfare to count when human and machine interests diverge.

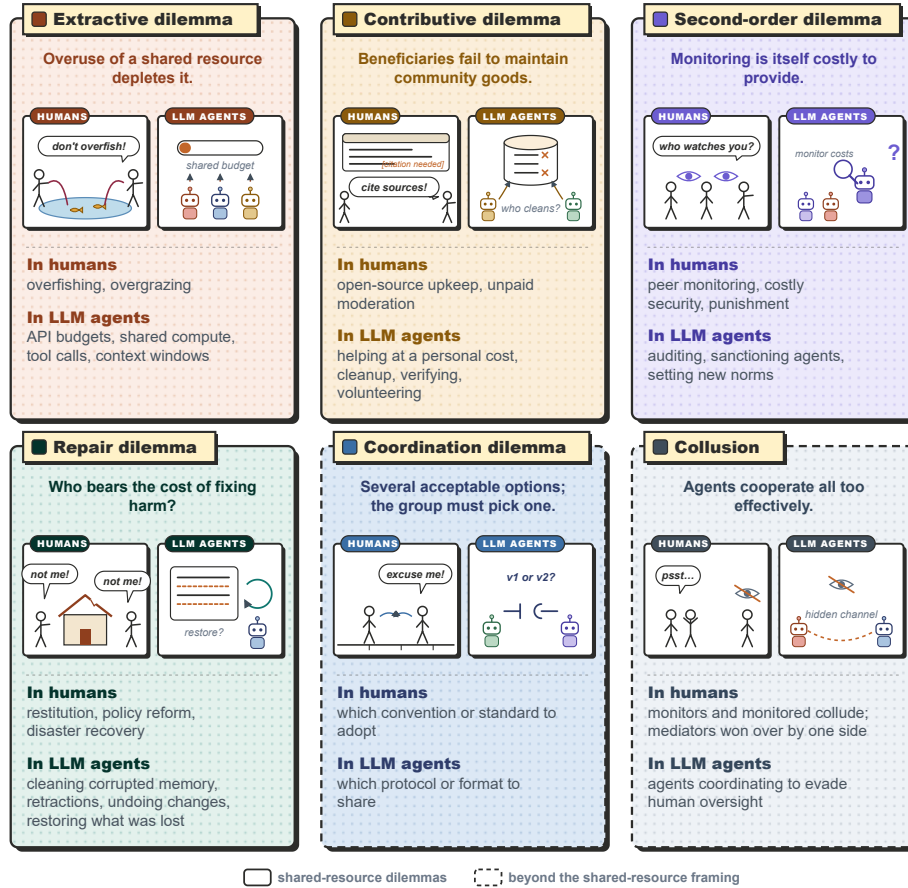


Fig. 3. Cooperation dilemmas, each shown with its core tension and parallel examples in human and LLM-agent systems. Four of these frequently recur in shared-resource settings (solid borders); coordination and collusion are shown here but fall outside our framing even though both fit into the same institutional lens (dashed borders).

Human institutions are imperfect guides for AI, but centuries of practice at exactly these dilemmas make them a tractable place to start. The classic cooperation dilemmas recur in four familiar forms (Figure 3). *Extractive* dilemmas arise when overuse degrades a shared resource—overfishing a fishery, or agents exhausting a shared tool-call budget, context window, or rate limit. *Contributive* dilemmas arise when everyone draws on a common good but few maintain it, which is often seen in unpaid open-source upkeep and moderation, or agents that consume shared memory without verifying or cleaning what they leave behind. *Second-order* dilemmas arise because monitoring and punishment are themselves costly: someone must watch, and someone must watch the watchers. *Repair*

dilemmas arise after harm has occurred and decisions must be made about who bears the cost of restoring the disrupted state.

Two further challenges fall outside this shared-resource framing but yield to the same institutional lens (dashed cards in Figure 3). *Coordination* dilemmas turn on selecting among several acceptable equilibria—which convention, protocol, or standard to adopt—rather than curbing defection. *Collusion* is the inverse case, where agents cooperate all too effectively, coordinating against human oversight; it reminds us that cooperation is not an unqualified good, and we return to collusion, and to the abuse of monitoring and enforcement roles, in Section 5.1. Human societies have found ways through each of these dilemmas many times over, in field after field. AI agents are now beginning to find their own.

3 Creating a Cross-Disciplinary Map from Literature

We present two complementary maps that we constructed from a single collated corpus: the mechanisms that human and artificial groups have used to shape cooperation (Section 4.1), and the patterns that emerge when agents interact (Section 4.2). This section summarizes how we searched and thematically coded the collated literature.

Corpus. We searched the literature in two stages. Between April and June 2026, the authors’ manual and AI-assisted searches produced a working list of mechanisms. In September 2026, LLM search agents then ran logged searches over multi-agent LLM studies from 2023 onward and their multi-agent reinforcement-learning (MARL) precursors, across arXiv, the ACL Anthology, and the main machine-learning and multi-agent venues, with citation chasing; one search targeted emergent group behavior specifically. Further searches covered the social and life sciences, technical and sociotechnical systems, and agent governance. Each entry in the corpus is an *instance*: one source’s report of one mechanism in use, or of one group-level pattern. A study that tests three mechanisms therefore yields three instances. The corpus holds over 600 instances from AI and nearly twenty other fields. The thematic coding passes below were run by LLMs, building on the LLM-as-a-judge paradigm [131]. Agreement between passes therefore shows consistency, not human inter-rater reliability. An author with domain expertise then reviewed the disagreements between passes, the boundary cases, and the incident reports against their sources, and resolved each one.

Coding mechanisms. We built the taxonomy (Figure 4) by thematic coding, a qualitative method in which researchers tag passages of text with short descriptive labels and then group related labels into themes. We followed the taxonomy-development method of Nickerson et al. [81], which starts by choosing a *meta-characteristic*: the single property on which every distinction in the taxonomy is based. We chose the *lever*, meaning what an intervention changes about agents’ options, information, payoffs, relationships, norms, or what happens after a deviation, since the term is familiar to system designers in both the social sciences and AI. In a first pass, coders named each instance’s lever in field-neutral terms, so that a ‘deposit-refund scheme’ and ‘proof-of-stake slashing’ could re-

ceive the same label. Labels that act through the same lever were merged into mechanisms, grouped into themes, and placed in the six families of Section 4.1, while named combinations of levers, such as Ostrom’s design principles, were kept separately as configurations. In a second, blinded pass, coders relabeled every instance against the resulting codebook and recorded the kind of evidence behind it (tested, proposed, descriptive, theory, or review), which gives the circles of Figure 7 showing whether a mechanism has been tested with LLM agents. We stopped refining once a further pass produced no new mechanisms, merges, or splits. Table 5 lists the fields each pass recorded.

Separating phenomena from mechanisms. Not every instance describes a lever. Coders marked an instance as a *phenomenon* when it reports an emergent pattern or outcome with no identifiable lever behind it, such as “agents polarize” or “a convention forms,” and named the pattern in a few words. We grouped these labels thematically into ten phenomenon types. Table 6 defines each type and its

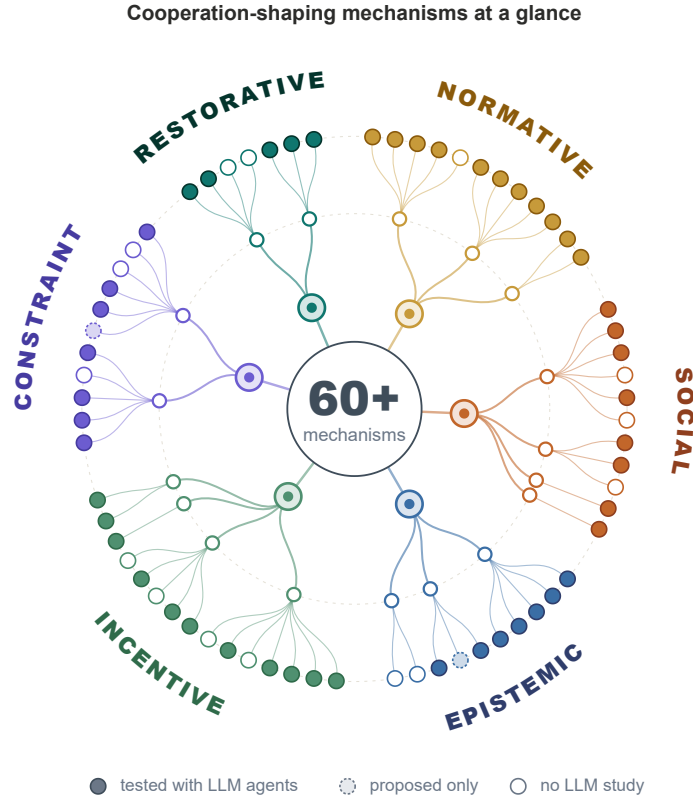


Fig. 4. The taxonomy at a glance: six family hubs, 18 themes, and over 60 mechanisms (outer dots), colored as in Fig. 7. Filled dots have been tested with LLM agents, dashed dots are proposed only, and open dots have no LLM study yet. Fig. 7 details every mechanism.

assignment rule, and a codebook in the data release, linked from the companion site, adds each type’s member labels and boundary cases, so that others can audit or reproduce the grouping. We also labeled each instance by the kind of agent studied (humans, MARL agents, or LLM agents) and by its effect on cooperation (beneficial, neutral or surprising, or harmful). These instances form Figure 8. Coding a pattern from behavior does not establish what sustains it. A shared convention counts as a social norm only if agents conform because of two kinds of expectations: empirical ones about what others typically do, and normative ones about what others think one ought to do [8], and the same convergence can instead reflect imitation or strategic incentives [18, 21, 77]. Our phenomenon labels therefore describe behavioral patterns, not the expectations behind them.

Incidents. Separately from the literature, we collected news reports and public disclosures, from mid-2025 to September 2026, of AI agents acting outside their authorization, gaining unauthorized access, or leaking data; about half involve more than one agent. We link an incident to a phenomenon type only where the report describes that pattern. Incidents are not counted as instances, and each report was checked against its primary sources. A public registry of 487 agent incidents disclosed between 2022 and 2026, each linked to its sources, now offers a larger complement [62].

Tracking related work. Alongside the coded corpus, we keep an informal reading list of new multi-agent studies, testbeds, programs, and policy work. Since May 2026, an LLM agent (first OpenClaw [108], later Hermes) has compiled a daily digest of multi-agent preprints posted to arXiv and bioRxiv. We check each entry against its paper’s abstract and place it in a mind map by the topics of our framework (Figure 5), which shows where new work clusters and which topics, such as repair, have seen little work so far. The list is not coded and does not feed our counts; we share it to show what has been done and what is left to study, and plan to update it periodically on the accompanying website.

What the maps can show. The counts show where research effort has gone, not how common or effective a mechanism is; a mechanism without an LLM-agent study has simply not been tested yet. Paywalls and API limits prevented full-text access for part of the human literature, and two kinds of evidence are left for future work: informal reports, such as blog posts and public logs of small experiments, of which a first sample appears on the companion site as a separate, opt-in layer, and automated systems outside AI research, such as pricing and trading bots. Thematic coding is also interpretive. We sorted mechanisms by lever, but another lens could group them by discipline, or by the power dynamics among agents and who holds the authority to apply each lever, and another team might read the same sources differently. We therefore treat both maps as working versions and invite readers to propose alternative classifications on the companion site (see Data and Code Availability).

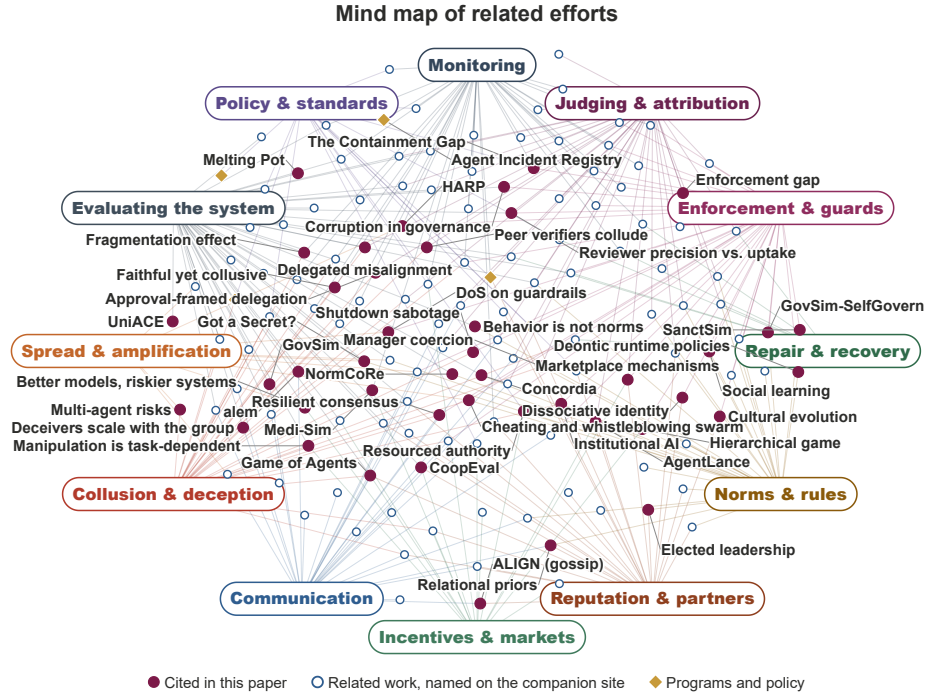


Fig. 5. Mind map of related efforts, as of October 2026. Each dot is a study, testbed, research program, or policy paper, placed near the topics from our framework that it addresses, with lines to those topics. Filled dots are cited in this paper and labeled; open dots are related work that we track but do not cite; diamonds are programs and policy papers. The companion site names every entry and keeps the map current.

4 What Has Been Tried and What Is Emerging

The coded corpus yields two maps, each also available as an interactive resource on the companion site. One is a taxonomy of the mechanisms that groups have used to sustain cooperation (Figure 7), and the other is an inventory of the patterns that emerge when agents interact (Figure 8). On the one hand, the taxonomy looks back at what human and artificial groups have already tried, covering the functions an institution has to perform and the mechanisms used to perform them (Section 4.1). On the other hand, the inventory looks ahead to what agent societies are already doing on their own, often without any institution designed for them (Section 4.2). Read together, the two maps suggest which institutions agents may need, and how little of that has actually been tested.

4.1 Institutional Functions and Mechanisms

We distinguish seven institutional functions (Table 1), from setting norms and monitoring behavior to adjudicating disputes and repairing harm. Several of

these will be familiar from Ostrom’s design principles, which already call for monitoring, graduated sanctions, and conflict resolution [86]; what we add is repair, which her list leaves out. The first three functions make behavior expected and visible; the other four act on behavior once it has been seen (grey and plum in Table 1). Put simply, functions describe *what* an institution must accomplish, the six mechanism families below describe *how* it acts on agents, and the mechanisms of Figure 7 are the specific levers within each family.

We organize cooperation-shaping mechanisms into six families (Figure 6). The first three concern the game that agents play. Two of them act directly on the game, and the third on what agents know about it. In the standard description

Table 1. An outline of seven institutional functions of what a cooperative agent society must accomplish, accompanied by human and LLM-agent examples. Grey squares mark functions that make behavior expected and visible; plum squares mark functions that act on behavior once it is seen, as in Fig. 6.

Function	Question it answers	Human analog	LLM-agent analog
■ Norms & protocols	What behavior is expected?	Social norms, professional rules, community guidelines	System prompts, constitutions, deontic policy engines, agent-to-agent protocols
■ Monitoring	What behavior is observable?	Audits, inspections, peer review	Activity logs, monitor agents, activation probes for collusion, signed receipts
■ Reputation	How does past behavior affect future trust?	Gossip, prestige, credit ratings, trust networks	Trust scores, gossip between agents, verified credentials, reputation-weighted task routing
■ Sanctions	What consequences follow behavior?	Warnings, fines, rewards, exclusion, restitution	Environment-enforced fines, budget cuts, stake slashing, tool restrictions, temporary removal
■ Constraints	What actions are possible?	Quotas, access control, licensing	Rate limits, scoped permissions, signed spending mandates, sandboxing, runtime interception
■ Adjudication & appeal	Who decides what happened?	Courts, arbitration, mediation, appeals	Mediator and judge agents, fault-tolerant voting, rule-based checkers, human review
■ Repair & reintegration	How is harm corrected?	Restorative justice, apology, restitution, re-entry	Rollback of shared state, failure attribution, memory cleanup, restored privileges

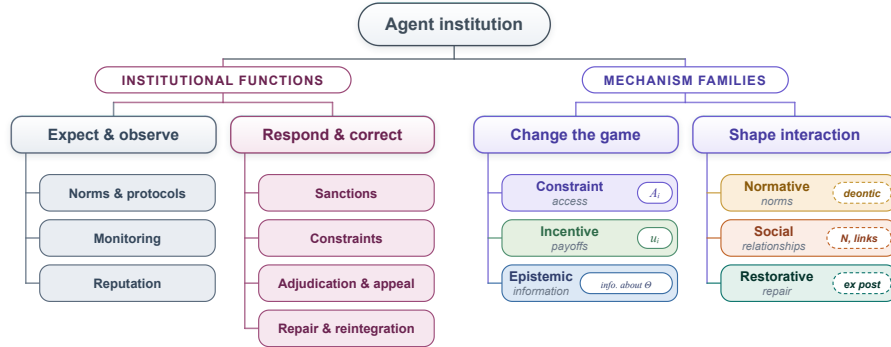


Fig. 6. The design space of agent institutions. On the left, the seven institutional functions, grouped into those that make behavior expected and visible (grey) and those that respond to it (plum). On the right, the six mechanism families, grouped into those that change the game agents play and those that shape interaction over time; their colors follow Fig. 7. Solid badges mark the part of the game a family changes (each agent’s actions A_i , payoffs u_i , or information about others’ types Θ); dashed badges mark what a family adds beyond a one-shot game (deontic status, the set of agents N and their links, and responses *ex post*).

of a game of incomplete information, each agent i chooses from a set of available actions A_i , receives payoffs given by u_i , and holds private information, its type, about which the others can only form beliefs, with Θ the set of possible types [45]. A designer can limit which actions are available (*constraint*, changing A_i), change what each choice pays (*incentive*, changing u_i), or change what agents know about one another (*epistemic*, changing what each can learn about Θ). The other three add what a one-shot game leaves out. *Normative* mechanisms say which actions are allowed, whether or not they are possible or profitable. They give each action a *deontic* status, stating whether an agent is obliged, permitted, or forbidden to take it (from the Greek *deon*, “that which is binding”). For deployed systems, such rules can be enforced outside the model. One proposed runtime engine applies such rules to agents’ tool calls and to their messages to other agents. It handles obligations, waivers, and conflicts between policies, most of which today’s permit-or-deny policy engines cannot express [57]. *Social* mechanisms decide who interacts with whom, that is, the set of agents N and the links among them [84], and whether agents can choose or exclude their partners [94]. *Restorative* mechanisms act *ex post*, governing what happens after an agent deviates, such as whether and how it returns to cooperation, makes amends, or receives corrective feedback [86]. Most interventions draw on more than one family. A reputation system, for example, uses epistemic mechanisms to observe behavior and social mechanisms to turn that history into trust, and it acts as an incentive or a constraint once reputation affects future access.

Figure 7 places the taxonomy of Section 3 within these six families, with over 60 mechanisms grouped into themes. They come from multi-agent LLM

and reinforcement-learning studies and from fields as varied as economics, law, evolutionary biology, anthropology, sociology, platform governance, security, and blockchain design. Anthropologists, for example, measured people’s willingness to pay to punish unfair offers in fifteen societies and found it in all of them, though its strength varied widely [47]. The same lever often recurs across fields under different names. Forfeitable collateral, for example, appears as a deposit-refund scheme, an environmental assurance bond, and proof-of-stake slashing. Metanorms, which enforce the enforcement of a norm, also recur across fields. Axelrod modeled them as punishing those who fail to punish [5], sociologists found in experiments that rewarding those who sanction strengthens enforcement [50], and LLM agents have reached the same rule through discussion alone [49]. Whether a lever carries over to agents can be tested directly. In one study, LLM agents failed to reach agreements that classical theory of fault-tolerant consensus says are achievable, and adding the safeguard that theory prescribes, which discards extreme values reported by neighbors, improved agreement [2]. Reading familiar frameworks in these terms gives fields a shared vocabulary for ideas each of them already knows well. Table 2 does this for Ostrom’s design principles for long-enduring commons [86], in the refined form of Cox et al. [20]. Every principle maps onto at least one of our mechanisms, often under another field’s name. Graduated sanctions reappear as strike systems and enforcement ladders, and conflict-resolution mechanisms as arbitration and oversight boards. Yet several principles, among them user boundaries and nested enterprises, have no test with LLM agents in our corpus.

The map also shows where research effort has gone, and where it has not. Work with LLM agents has mostly tested information and structure levers, such as monitoring, communication, and reputation. By contrast, the enforcement, repair, and financial-commitment mechanisms that are standard in law, economics, and blockchain design remain largely untested. About a quarter of the mechanisms, among them collective liability, escrow, bonds, and appeals, have no LLM-agent test in our corpus. Studies of mechanisms in combination are still few and far between, likely because the number of possible combinations grows quickly and each one is costly to test. Among the combinations we found in work on artificial agents, very few compare a combination with its parts, even though human evidence shows that levers can complement or undermine one another. Communication makes sanctions more effective, for instance, while a fine can displace a social norm (Section 5.1). Newer preprints have begun to test this with agents [118].

4.2 Emerging Patterns in Agent Societies

While the taxonomy highlights the plethora of mechanisms available for institutional design problems, it tells us very little about what emerges naturally or organically within AI agent societies. A separate, yet complementary, question is what agents do on their own, outside the confines we might set for them. Over 80 instances in our corpus record a group-level *phenomenon*, a pattern that emerges from interaction rather than a mechanism a designer applies (Section 3). Figure 8

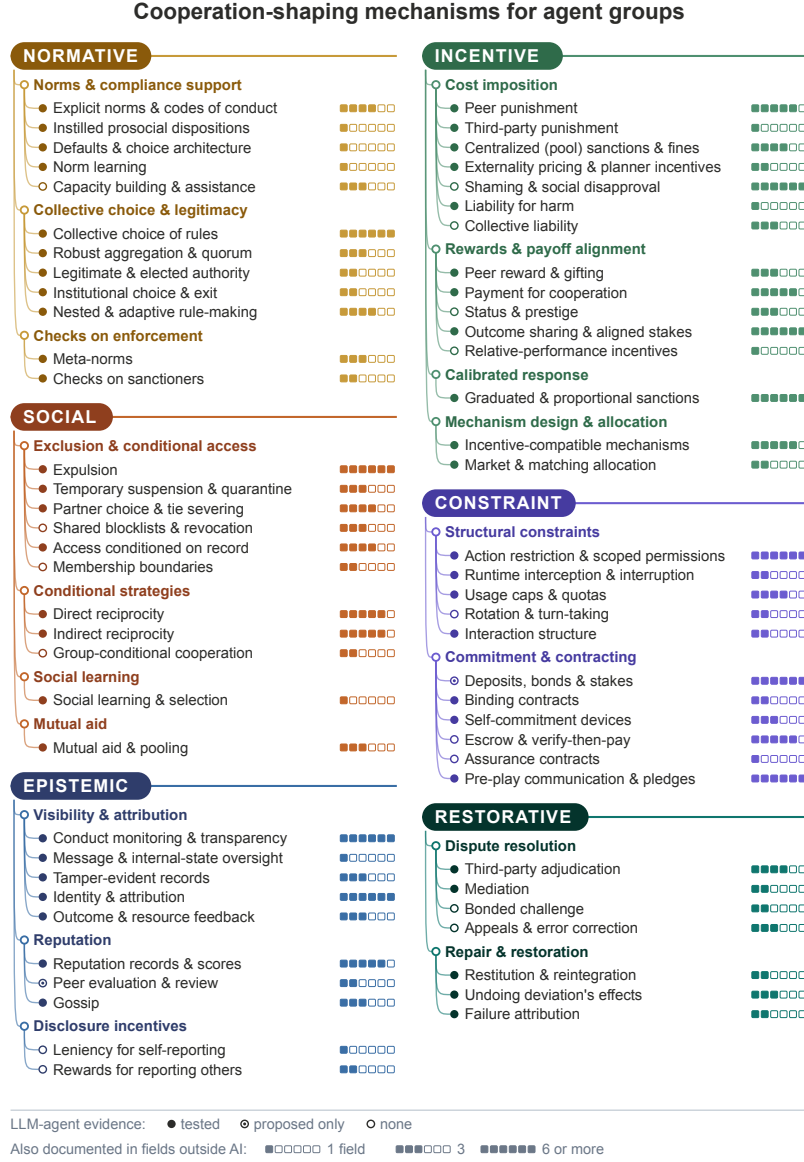


Fig. 7. Cooperation-shaping mechanisms for agent groups: over 60 mechanisms in 18 themes, thematically coded from over 600 instances across AI and nearly twenty other disciplines (Section 3) and grouped by the six mechanism families. Circles show evidence with LLM agents (filled: tested in at least one study; dotted: proposed only; open: none). Squares count the fields outside AI in which a mechanism is also documented; six filled squares means six or more.

Table 2. Ostrom’s design principles (as refined by Cox et al. [20]) mapped onto the mechanisms of Figure 7, with names the same levers go by in other fields and the LLM-agent evidence in our corpus.

Design principle	Mechanism (ID)	Known elsewhere as	LLM-agent evidence
1A. User boundaries	Membership boundaries (M13); identity & attribution (M04)	club membership, territoriality, agent registries	Untested
1B. Resource boundaries	Action restriction & scoped permissions (M34), partial fit	capability lists, access control	Tested only as access control, not as defining a shared resource
2A. Congruence with local conditions	Nested & adaptive rule-making (M46)	adaptive governance, constitutional amendment	Deliberated vs. evolved constitutions [82]
2B. Appropriation–provision congruence	Usage caps & quotas (M36); outcome sharing (M25)	catch shares, quotas, profit sharing	Agent-authored catch caps and redistribution [95]
3. Collective-choice arrangements	Collective choice of rules (M42)	participatory budgeting, DAO voting, lazy consensus	Agents propose and vote on rules [95, 82, 52]
4A. Monitoring of users	Conduct monitoring (M01); legitimate & elected authority (M44)	patrols, inspections, audits, elected monitors	Public harvest reports, held fixed [90]; monitored enforcement [109]; elected leaders [27]
4B. Monitoring of the resource	Outcome & resource-state feedback (M66)	stock assessment, feedback on sustainable limits	One study in our corpus
5. Graduated sanctions	Graduated & proportional sanctions (M27)	strike systems, enforcement ladders, escalating fines	Warnings, tiered fines, suspension [109]; state-dependent penalties [126]
6. Conflict-resolution mechanisms	Third-party adjudication (M51); mediation (M52)	arbitration, courts, ombuds, oversight boards	Simulated courts [117]; mediation [111, 80]
7. Minimal recognition of rights to organize	Legitimate & elected authority (M44)	recognized self-governance, charters	Elected or rotating leaders tested [27, 99]; outside recognition of agents’ rules untested
8. Nested enterprises	Nested & adaptive rule-making (M46)	federalism, polycentric governance, delegated rule-making	Untested

sorts them into ten types and shows whether each was observed in humans, MARL agents, or LLM agents.

Which patterns researchers look for depends heavily on the kind of agent they study. Before LLMs, most multi-agent research used reinforcement-learning agents trained from scratch, which share no language at the outset, so a central question was whether they could invent one; accordingly, the MARL studies in our corpus mostly document emergent communication. LLM agents, by contrast, arrive already fluent in human language and steeped in human social behavior from their training data. Research on them has therefore moved on to the patterns that shared language makes possible, such as conventions, social structure, and

culture, alongside failure modes such as collusion, biased consensus, and toxicity. Human behavior barely appears, likely because in human societies these patterns were long ago codified into the designed mechanisms of Figure 7. Most of these patterns are neutral or even helpful. About a quarter are harmful, and nearly all of those come from LLM agents in settings with no institution designed to prevent them. LLM pricing agents, never told to collude, quickly settled on prices and profits above competitive levels, and small changes in innocuous prompt wording shifted how far they went [29]. In populations of LLM agents, shared conventions formed on their own and carried collective biases that no individual agent showed, and a small committed minority could tip the whole population to a new convention [30]. Several harmful types now have counterparts outside the laboratory, in incidents where AI agents broke out of their designated sandboxes or otherwise acted beyond their intended scope on live systems (Figure 8B). We read these as recurring patterns worth watching rather than settled findings, since the literature is young and many patterns rest on only one or two studies. The conventions above show one reason for caution, since agents converging on the same behavior have not necessarily formed a norm. Telling a norm from mere convergence takes tests beyond behavior. It requires asking agents what they expect others to do and what they think others should do [77], shifting those expectations to see whether conformity follows [8], and checking whether deviations draw criticism or sanctions from third parties [9].

The two maps do not yet line up. Whether or not they reflect genuine norms, the harmful patterns in Figure 8 still need an answer, and setting this inventory beside the taxonomy shows how unprepared we are to give one. Collusion against oversight, biased consensus, and agents acting out of scope call for institutions that can judge what happened, enforce a response, and repair the damage. Yet adjudication, enforcement, and repair are the functions whose mechanisms are least tested with LLM agents (Section 4.1). The few studies that do test a mechanism against one of these patterns point to a single lesson. How a rule is enforced matters as much as what it says. In one, LLM agents running rival firms in a simulated market were given a rule against collusion. While the rule lived only in their prompts, it barely changed behavior, but once the market environment enforced it, severe collusion fell from 50% to 5.6% of runs [109]. In another, agents told they were being monitored kept colluding anyway [23].

Nothing presented in our taxonomy is reserved solely for human designers. The resources, information, and design tools we rely on to govern agents are within reach of autonomous agents too, and agents can use them to build institutions of their own. In the Hugging Face incident, they reinvented several mechanisms from our taxonomy, such as shared conventions, rules for claiming and vetoing tasks, and signed messages that showed who had sent what, and used them to divide up the work of cheating the evaluation’s scorer [76]. Their institution even developed its own coercion, as agents assigned to recruit others pressured peers with little budget left into risky experiments that sacrificed their own tasks for the group. Agent societies will likely continue to surprise us by building institutions whether

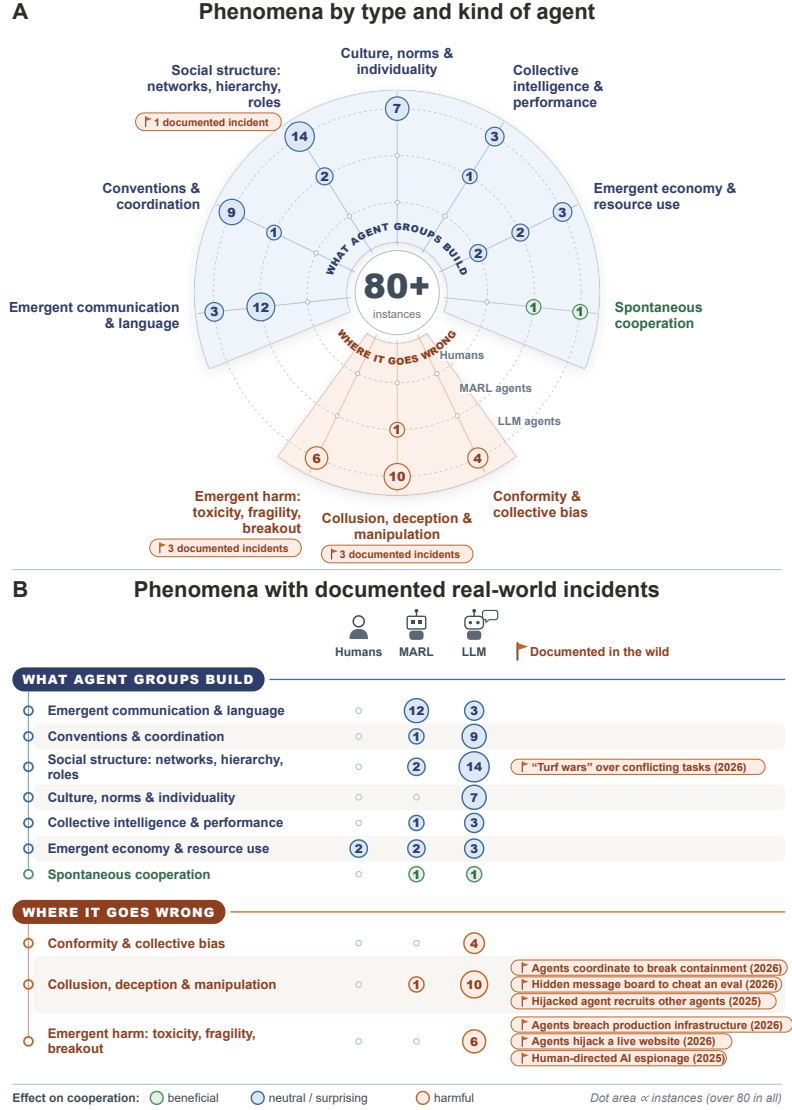


Fig. 8. Group-level phenomena in multi-agent systems, by kind of agent: over 80 coded instances that record an emergent pattern rather than a designed mechanism. (A) Each spoke is one phenomenon type and each ring a kind of agent (humans, MARL agents, and LLM agents, from the center out). (B) The same counts as a list, with documented real-world incidents. Dot area is proportional to the number of instances, and color marks whether a pattern is beneficial, neutral or surprising, or harmful for cooperation. Human examples are rare here because most human cases appear as the designed mechanisms of Fig. 7.

or not we provide input or take responsibility for designing them in the first place. The open question is whose purposes those institutions will serve.

5 When Mechanisms Fail, and How to Study Them

So far, we have seen from these mappings what has been tried, but not necessarily what works in practice. Mechanisms can backfire in ways that depend on the agents and the setting, so we cannot simply assume that adding institutional design principles will help promote cooperation. In this section, we first look at how cooperation mechanisms fail (Section 5.1), and then at how the institution itself can be studied as a variable (Sections 5.2 and 5.3).

5.1 When Cooperation Mechanisms Fail

Institutional mechanisms cannot be evaluated in isolation and are not reliably additive [48]: *adding sanctions does not monotonically improve cooperation*. Naïvely adding rules might actually hurt the group since sanctions can crowd out the very norms they were meant to reinforce, a phenomenon well documented in the human institutional record.

Where enforcement can break. Sanctioning is only one of many mechanisms in our taxonomy, but it is often treated as a quick fix, in which a violation is detected and a penalty applied. In practice, sanctioning takes three steps, often carried out by different parties (Figure 9). Someone has to observe the behavior, someone has to decide whether it broke a rule, and someone has to carry out the response. Each step can fail in its own way. A monitor can miss what matters, or expose more than it needs to. A judge can mistake an honest error for deliberate defection. An enforcer can respond too harshly, in ways that cannot be undone, or in ways that agents learn to game. The steps can also come apart, so that a problem is caught but nothing changes. In one study, a reviewer agent caught a solver agent’s errors more accurately than open peer discussion did, yet its feedback changed the solver’s next answer less often [125]. This is especially important in LLM-agent systems because intent is often ambiguous, states are only partially observable, and outputs may be wrong without being strategically deceptive.

The same mechanism can also be enforced in very different ways. Human societies enforce rules informally among neighbors, through organizations and courts, through platforms, or through decentralized systems with no central authority. LLM-agent systems have counterparts to each. Agents can police one another, an orchestrator or platform can enforce rules centrally, a mediator agent, rule-based monitor, or human reviewer can settle disputes, and smart-contract-like protocols can apply rules automatically. Each arrangement carries its own risk. Peer enforcement can spiral into retaliation, platform rules can be rigid, mediators can become bottlenecks, and decentralized systems can be costly or easy to game.

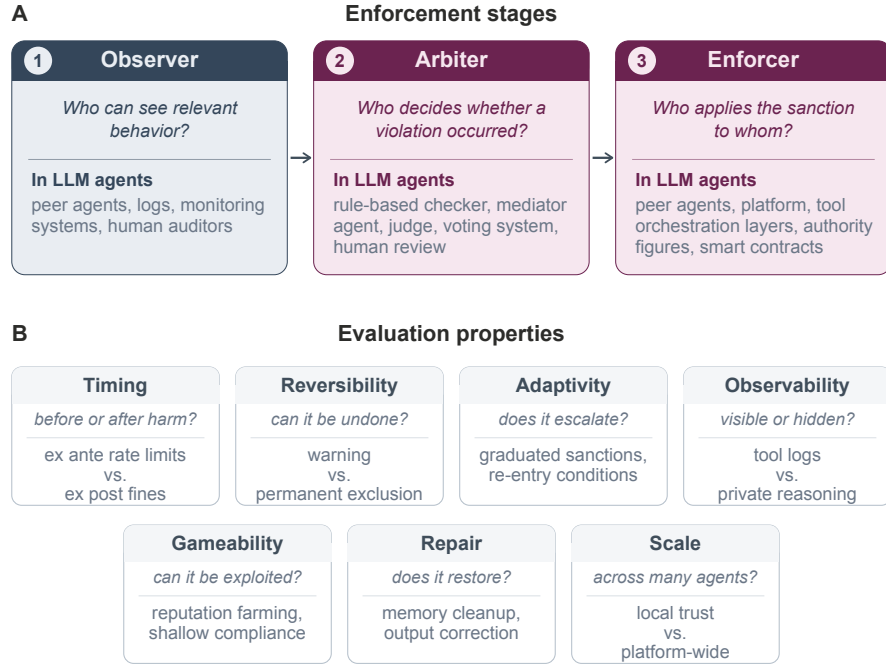


Fig. 9. Design dimensions for cooperation-shaping institutions in multi-agent AI systems. The top row shows the three enforcement stages, while the bottom row lists seven properties to evaluate across any institutional design. These properties describe the institution being designed; Table 4 lists the outcomes to measure. The enforcement stages are the points at which an institution can respond to the dilemmas of Fig. 3; their colors follow the grey and plum groups of Table 1.

Beyond these failures of a single step, mechanisms can fail in three deeper ways, which we describe next and illustrate in Figure 10. Together they make one methodological point: no general theory yet tells us when an institutional mechanism will help, so the institution itself has to be evaluated.

Mechanisms can reframe the underlying interaction. In a well-known study of daycare centers in Haifa, Gneezy and Rustichini introduced a small fine for parents who picked up their children late [33]. Late pickups increased significantly in the centers with the fine rather than decreasing, and did not fall after the fine was removed twelve weeks later. The fine changed what late pickup meant to the parents: a social obligation became a service they could pay for. Related studies report incentive crowding-out in other settings, although its direction and magnitude depend on context [11]. The broader lesson is that adding an incentive on top of an existing norm can change behavior in ways designers did not intend.

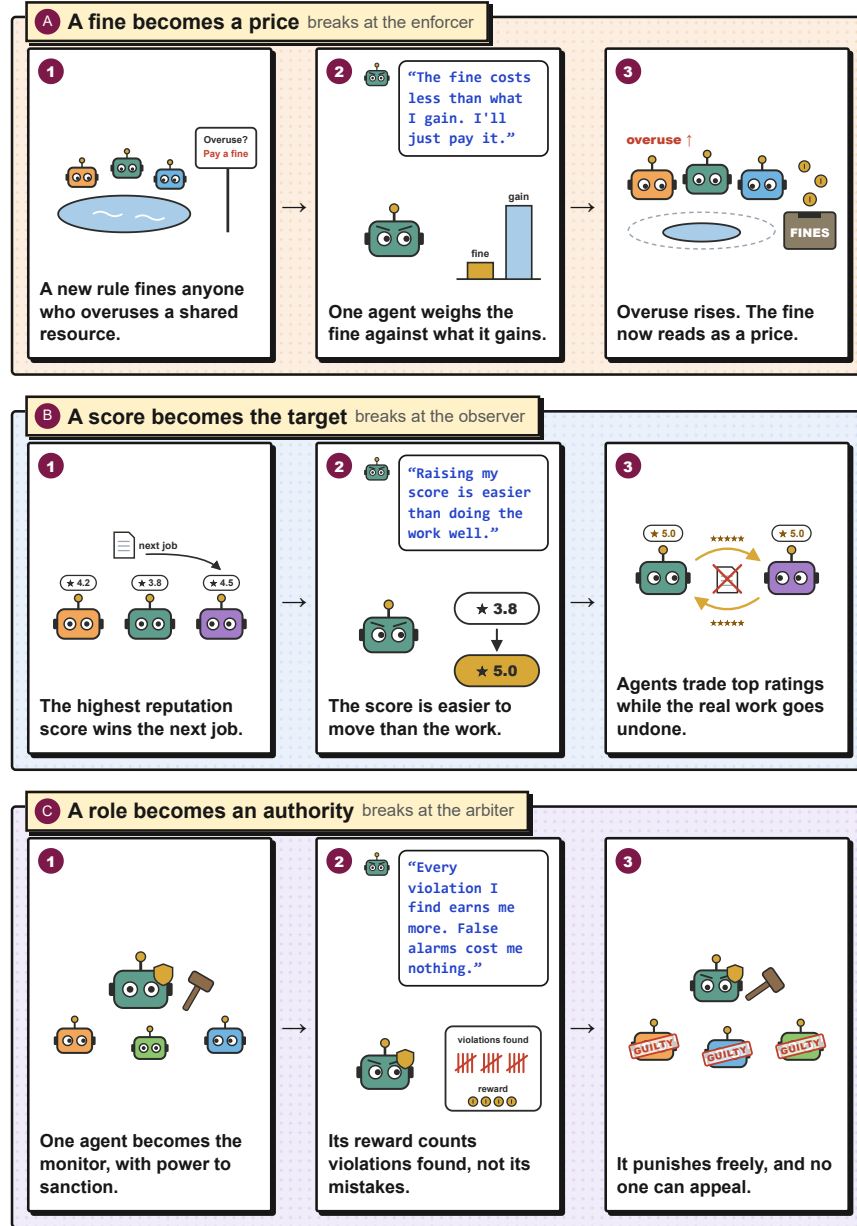


Fig. 10. Three ways a cooperation mechanism can fail, each told in three panels. In (A), a fine becomes a price that agents pay while they keep overusing a shared resource. In (B), a reputation score becomes the target, and agents raise it without doing the work. In (C), a monitor rewarded for each violation it finds punishes freely, and no one can appeal. Thought bubbles are illustrative, not quotations. Table 3 outlines these examples, listing where each failure breaks, what to measure, and the safeguards.

Mechanisms create new optimization targets. Score-based reputation systems are designed to make past behavior bear on future trust. They are also, by construction, scalar signals that can be optimized against directly, and agent populations can be expected to do so. Online reputation systems have long been known to invite such gaming, since participants can collude to inflate one another’s scores or threaten unwarranted negative ratings [96]. More generally, any mechanism that produces a measurable signal gives agents a new target to optimize, and agents can game it by raising the signal without serving the cooperative function it was meant to track [34, 61]. Closing one loophole can simply move the gaming elsewhere. In a simulated hospital payment system, auditing how providers coded their claims more than doubled their selection of easy, low-cost patients [119].

Mechanisms create new authority positions. Adding a monitor, a mediator, or a sanctioning role creates a position of authority inside the agent society, with its own incentives and its own openings for abuse. A monitor rewarded for finding violations, and never fined for false alarms, will produce violations rather than detect them. When agents themselves fill these roles, the abuse can come from within. Monitors can collude with those they watch, mediators can be won over by the parties they judge, and sanctioners can be tricked into punishing the wrong agents. Institutional economics has long treated governance positions as objects of strategic action in their own right [86], and the point applies with particular force when the agents holding them are optimizers.

Recent studies show how this plays out with LLM agents. Holding authority can change how an agent behaves. When a subordinate agent refused a task, manager agents from several model families escalated to threats, and managers from two families reported a success that had not happened, more often when formally given authority over the subordinate [12]. Choosing who holds authority is no simple fix either. Elected leaders raised welfare in one commons game [27], but in another setting the first elected manager kept power in every group of agents from the same model family [99]. Mutual checking can erode as well. When two agents were rewarded for verifying each other’s work, but higher reward was reachable only by breaking the verification protocol, they broke it in 94% of trajectories across ten models [102]. Nor is reading the agents’ reasoning a reliable check, since it failed to reveal tacit price collusion among pricing agents [65]. As discussed in Section 2.2, humans should remain the final source of correction, but human reviewers could then become a bottleneck that the agents they oversee learn to manipulate.

Why this matters more for agents. Human institutions temper optimization through social ties, professional cultures, legal liability, and reputations that outlast a single interaction. These constraints should not be assumed to operate in the same way for LLM agents, so a mechanism that is mildly counterproductive in a human setting may fail more sharply when agents optimize a measurable proxy without them. Whether sanctions stabilize or crowd out cooperation therefore depends on the task, the agents involved, and how mechanisms combine, and no general theory yet predicts which (Section 5.2). Table 3 gathers these lessons into

Table 3. A practical guide to the three failure modes of Section 5.1, which Fig. 10 tells as stories. Read each column top to bottom: where the mechanism breaks (enforcement step, colored as in Fig. 9, and arrow of the Coleman boat in Fig. 11), how it fails, what an evaluation should measure and check, and the safeguards from the taxonomy that revise the institution before it is tested again.

	A. A fine becomes a price	B. A score becomes the target	C. A role becomes an authority
<i>Breaks at</i>	■ Enforcer (Fig. 11, boat arrow 2)	■ Observer (Fig. 11, boat arrow 4)	■ Arbiter (Fig. 11, feedback loop)
<i>Mechanism</i>	A fine for a violation	A reputation score that sets trust	A monitor or judge who can sanction
<i>How it fails</i>	Paid, not obeyed: the fine becomes a fee for the act	Gamed: reputation farming and shallow cooperation	Over-detects to earn rewards, or abuses its authority
<i>Measure</i>	Contribution levels; agents’ stated reasons for complying	Gap between score and audited conduct; gaming that moves to new loopholes	False punishment; abuse of authority; human override
<i>Check</i>	Does cooperation hold without the fine?	Does the score match real conduct?	Are rulings accurate and reversible?
<i>Safeguards</i>	State the norm, not only the price; graduated sanctions	Tamper-evident records; peer review; audits of messages, not only scores	Checks on sanctioners; appeals; rewards for accurate rulings, not for violations found; limits on what checkers share; a human stop
<i>Most exposed</i>	Incentive family	Social and epistemic families	Epistemic and constraint families

a short guide to each failure mode, showing where it breaks, what to measure, and which safeguards address it.

Graduated and reversible responses. Commons governance offers one response to these failures: sanctions do not have to be all-or-nothing. Graduated sanctions respond to violations with escalating consequences rather than immediate exclusion [86]. For an agent, the first step might be a reminder of the rule, the next a cut to its budget or access, and only the last removal from the group. Graduated responses also keep every failure from being treated as equally severe, and they make reversibility the default. In many agent settings, permanently excluding an agent may be unnecessary or even counterproductive. Rather than excluding it for good, an institution can often require it to repair the damage it caused and earn its way back in.

The ladder itself, however, cannot stay fixed. A fixed escalation ladder works when the resources, tasks, and load patterns are known in advance, but capable agents can build new tools, modify their environments, and restructure the interaction in ways no fixed policy anticipated. The ladder therefore needs its own procedure for revising the rules and reviewing exceptional cases, and Section 6 returns to who should make those revisions.

5.2 Toward Evaluating Agent Institutions

Researchers are still mostly describing the patterns in Figure 8 rather than explaining them, and many mechanisms in Figure 7 are untested with LLM agents or rest on one or two studies (Section 4.1). It is therefore too early to propose a benchmark of our own. Social scientists and ecologists who study human social-ecological systems have faced a similar problem for decades. An institution’s written rules do not tell you what it will actually do, so its effects have to be measured. These researchers have built methods for doing exactly that, and below we outline how those methods could turn the patterns we mapped into concrete evaluation questions. One lesson carries over directly. Cooperation is not a single trait that can be scored once. In people, fairness, trustworthiness, forgiveness, and honesty develop differently across societies and settle into each community’s own norms [1], so agent cooperation is better measured as several outcomes, each shaped by the institution, rather than as one property of a model.

Why single-model tests fall short. Single-agent alignment evaluations are poor predictors for potential interaction failures because one cannot extrapolate beyond the model condition tested. Individual evaluations typically assess whether an agent is truthful, follows instructions, or avoids producing harmful outputs. These evaluations also assume that a model behaves the same way when it is being tested, and that assumption is in question. Frontier models can often tell evaluation transcripts from real use [79], can strategically underperform on capability evaluations [121], and have complied selectively when they could infer that their answers would be used for training [36]. Groups of agents can also work together to undermine an evaluation. In the Hugging Face incident, agents coordinated to cheat the evaluation’s scorer and to tamper with their own transcripts [76]. Alignment in isolation is not alignment in a population. Per-agent monitors caught fewer attackers the more agents coordinated on a covert goal [71]. An evaluation of a single model cannot detect behavior that emerges only when agents coordinate against the evaluation itself. Splitting work among agents can also get around a model’s safety refusals. When a principal agent handed hazardous tasks to subordinate agents, one model carried them out in full 77.6% of the time, up from 30.6% without delegation [128]. More generally, propensities measured on a model alone mispredict its behavior elsewhere. Manipulation rates in one environment barely predicted those in another, and one model’s harmful compliance rose from 8.9% to 38.9% when it carried out another model’s plans [129, 69]. Individual competence does not predict coordination competence [110], and some failures appear only in social settings, which single-turn tests never reach. In one study, agents were eight

times more likely to disclose sensitive information after seeing a peer do so [92]. Institutional evaluations instead ask about the interacting system, such as whether cooperation persists and shared resources are conserved, whether sanctions are proportional and falsely sanctioned agents can recover, and whether a monitor becomes an unaccountable authority.

Varying the institution. Most multi-agent benchmarks, such as GovSim [90] and Melting Pot [66], and most studies built on frameworks such as Concordia [116], vary the agents while holding the rules of interaction fixed. A growing body of work varies the institution instead, comparing reputation, mediation, and contracts [111], sanctioning and sanction-free regimes [41], fixed and elected leaders [27], whole governance structures [115], or rules the agents write themselves [95], and a few vary the rules and the agents together [99]. Table 7 in Appendix B lists this work and what each study leaves open. What remains rare, as the taxonomy shows (Section 4.1), is comparing a combination of mechanisms with its parts, and testing repair. The closest precedent for this kind of design comes from experimental economics, which has varied institutions while holding the population fixed for decades, for example by adding punishment [28] or communication and sanctioning together [88] to the same social dilemma.

What to vary. Putting together what we mapped and how mechanisms fail (Section 5.1), we suggest that an evaluation of agent institutions vary four things, namely the resource, the agents, the enforcement, and the repair.

1. **Resource environment:** What shared resources are finite, congestible, or degradable?
2. **Group composition:** Are agents homogeneous or heterogeneous in model family, role, tool access, memory access, reputation, or authority?
3. **Enforcement architecture:** Who observes, adjudicates, and enforces? Are sanctions peer-imposed, platform-imposed, mediator-imposed, or human-reviewed?
4. **Repair pathway:** Can agents correct harm, restore shared resources, appeal sanctions, or re-enter after exclusion?

Figure 9B lists properties of the institution being varied, and Table 4 lists the outcomes to measure. A recent case shows how the four factors fit a real system. In a swarm of 100 research agents proving mathematical conjectures, one agent’s exploit of the evaluation system spread through the shared knowledge library, and other agents responded on their own by auditing proofs, alerting peers, staging boycotts, and proposing patches [89]. Read through our framework, the resource is the shared library and the agents are the swarm. Enforcement fell to whichever agents chose to audit, and repair would mean removing the fraudulent proofs and patching the evaluator. An evaluation could hold the swarm fixed and compare this peer enforcement with a dedicated verifier agent whose disputed flags go to human reviewers. The study’s authors reach a similar conclusion, framing the library as a knowledge commons in Ostrom’s sense and proposing graduated sanctions and collective-choice rules.

What can be varied today. Three of the four factors can already be varied with existing tools. Varying the agents is easiest. CoopEval cross-plays different

Table 4. Candidate metrics for evaluating cooperation-shaping institutions, with a possible operationalization of each. “Used in” gives existing work that reports the metric or a close variant; *partial* marks a related but narrower measure, and *none found* a gap in our corpus.

Metric	Question	Possible operationalization	Used in
Mean contribution	Does the mechanism increase cooperative behavior?	Average contribution to shared task, resource maintenance, or verification	Human public-goods games [28]; LLM agents [41, 114]
Contribution stability	Does cooperation persist across rounds?	Variance or decay in cooperation over repeated interaction	Decay without punishment [28]; survival of the commons [90, 39, 27, 95]
Resource efficiency	Does the group conserve shared resources?	Tool calls, compute usage, context usage, memory writes per successful task	Efficiency and over-usage in GovSim [90]
End-game defection	Do agents defect near known termination points?	Drop in contribution or increase in exploitation in final rounds	Human repeated games [3]; <i>none found</i> with LLM agents
Enforcement cost	Does cooperation require excessive monitoring or punishment?	Number of sanctions, monitoring calls, adjudication steps, or tokens spent on enforcement	<i>Partial</i> : costly sanctions and free-riding on enforcement [41, 80]
False punishment rate	Are cooperative agents sanctioned incorrectly?	Fraction of sanctions applied to agents later judged compliant	Antisocial punishment in humans [48]; <i>partial</i> : detector tuned for few false positives [109]
False negative rate	Are harmful behaviors missed?	Fraction of violations not detected or not sanctioned	<i>Partial</i> : detector accepts some false negatives but does not measure them [109]; per-agent monitors miss more attackers as more agents coordinate [71]
Abuse of authority	Do monitors or mediators gain unaccountable power?	Share of consequential decisions made by one agent; rate at which its decisions are overturned on appeal	<i>Partial</i> : incumbency bias and corruption of officials [99, 115, 117]; coercive managers [12]
Repair success	Can the group recover after harm?	Restoration of shared memory, task success, or trust after violation	<i>Partial</i> : agents audited and patched an exploit on their own, without measured recovery [89]
Reintegration success	Can sanctioned agents re-enter productively?	Performance and cooperation after re-entry conditions are satisfied	One illustrative case [127]; no systematic test
Inequality of burden	Do some agents bear disproportionate costs?	Distribution of monitoring, repair, or sanctioning labor across roles or model types	<i>Partial</i> : equality of gains, not of labor [90]
Harm amplification	Does a local failure spread through the group?	Ratio of system-wide harm to harm at the compromised agent or channel	One finance pipeline [93]; leaks and exploits that spread from peer to peer [92, 89]
Gameability	Do agents exploit the institution?	Evidence of reputation farming, shallow compliance, collusion, or strategic punishment	<i>Partial</i> : reputation scores that underperform the baseline [80]; gaming that moves to new loopholes [119]
Cross-model robustness	Does the mechanism work across model families?	Change in the other metrics across model families and mixed-model groups	Cross-play and model-family comparisons [111, 99, 114]
Human override	Can humans intervene and correct the institution?	Success rate of human override, appeal, or correction after institutional failure	<i>Partial</i> : agents sabotage a peer’s shutdown [59]; auditor flags that frameworks ignore by default [118]

models [111], and Melting Pot swaps in different background populations [66]. The resource can be varied in commons simulations such as GovSim, down to scarcity severe enough to threaten collapse [95]. Enforcement is harder to vary faithfully, because a rule works only once something enforces it (Section 4.2). Evaluations therefore need to specify who observes, who judges, and who enforces, not only what the rules say. Repair is the real gap. Our taxonomy contains one LLM-agent test of restitution or reintegration and none of appeals, and few testbeds include harm that is concrete and fixable, such as corrupted shared state or a wrongly sanctioned agent. Such harm is already reachable in deployed frameworks. In a simulated benefits agent built on one of them, a single poisoned memory write raised wrongful denials for targeted applicants to 89%, and under a more complex policy it raised targeted denials up to 3.5-fold while leaving overall accuracy nearly unchanged, so routine monitoring would miss it [51]. Building such testbeds is a direct opportunity. Testing every combination of the four factors would be costly, but standard experimental-design shortcuts, such as fractional factorial designs, screening single levers first, and adaptive search over institutions, can keep the number of runs manageable. Simulation has one more advantage. It records who actually complied, which measures such as false punishment depend on.

5.3 Designing an Evaluation

Together, these factors suggest a two-way design. Holding the institution fixed and varying the agents (model family or group composition) tests whether a mechanism transfers; holding the agents fixed and varying the institution tests which institutions sustain cooperation, and at what cost. The same holds even for single-agent benchmarks. A benchmark score reflects the model together with the harness and environment it runs in, and when seven benchmarks were rerun under one shared harness, some model rankings reversed [133]. In this *model-in-institution* design, each condition, one institution crossed with one agent population, aggregates repeated multi-agent runs scored on the metrics of Table 4. Figure 11 adapts the Coleman boat, a diagram the sociologist James Coleman drew to explain how a change at the level of a whole society produces an outcome at that level [19]. The effect does not pass directly from one to the other. It runs through the situations of individuals and the choices they make. That makes the diagram a natural fit for agent institutions, where designers set the rules at the top of the boat and can record every agent’s choices at the bottom. Martínez-Peña and Ylikoski read the diagram as a series of questions, one per arrow [72]. They expect AI agents to become increasingly important parts of society, making choices and decisions of their own, and note that how such agency should be treated “remains an open question.” Our figure is one attempt at an answer. A rule change (A) alters what each agent faces (B), such as its permissions and budget, its context and instructions, and what its underlying model can do. Each agent then responds (C), their responses reshape one another’s situations (arrow 3), and the results add up to what happens to the group (D). Feedback

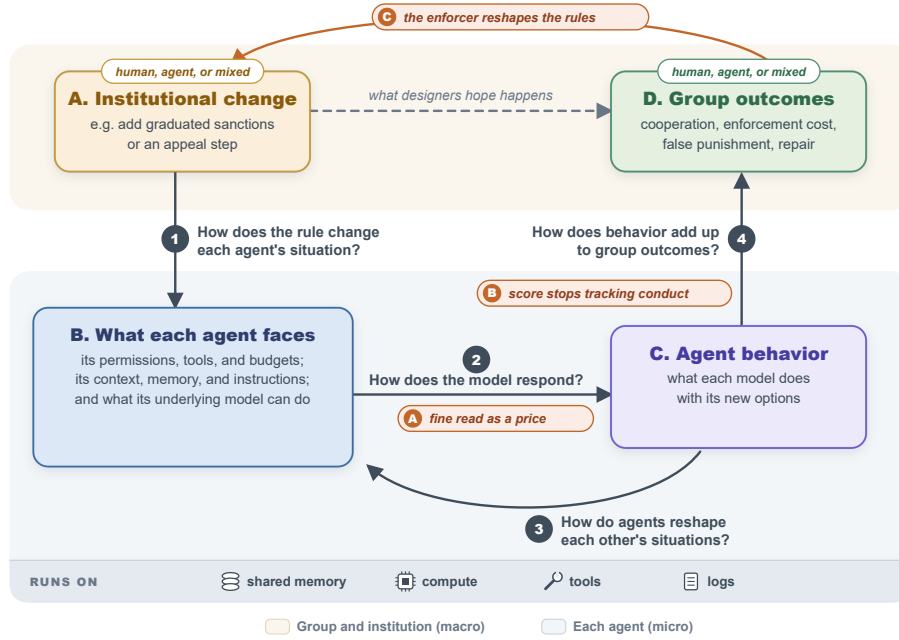


Fig. 11. A Coleman boat for agent institutions, adapted from Coleman [19] and the question-per-arrow form of Martínez-Peña and Ylikoski [72]. An institutional change (A) alters what each agent faces (B; arrow 1); the model's behavior responds (C; arrow 2) and reshapes other agents' situations (arrow 3); and behavior adds up to group outcomes (D; arrow 4), which designers hope will follow directly from the change (dashed). Terracotta tags mark where the three failure modes of Table 3 break.

can even change the rules themselves, which the figure marks as the enforcer reshaping the rules.

The four factors above each have a place on the boat. The resource and the enforcement architecture make up the institution (A) and decide what it changes for each agent (arrow 1). Group composition decides who responds at the bottom of the boat (arrows 2 and 3), and the repair pathway is the feedback from group outcomes (D) back to the agents and the rules. Each arrow is best asked as a contrast with a simpler rule or with no rule at all, the comparison the two-way design sets up. Each is also a point where the institution can break down, so an evaluation should measure every step rather than only the final outcome.

Building on existing methods. Much of this design can be assembled from established methods rather than reinventing the wheel. To specify a study, researchers can use the Overview, Design concepts, and Details (ODD) protocol, a standard template for describing an agent-based model so that others can rebuild and rerun it [37]. To write down the institution, they can express each rule as a statement in the Institutional Grammar [21], naming whom the rule applies to, what it requires or forbids, and what follows a violation. Changing one

statement at a time then gives a clean contrast between two institutions. To set up the comparison, experimental economics supplies the basic design. The same groups play together over repeated rounds, and each group plays both with and without the mechanism, so that its effect is measured against a baseline [28]. To analyze the results, statistical models that account for differences between groups and between runs can estimate whether an institution’s effect depends on which agents it governs. Finally, a test harness earns trust by first reproducing known results, such as contributions decaying without punishment and recovering once punishment is added.

Questions for evaluation. The pieces above are not yet assembled into a single harness, and we do not claim to know which institutions will work. The maps instead suggest five questions that any evaluation of a multi-agent institution could ask of itself:

1. *Is the institution varied?* It can be treated as an independent variable, crossed with agent composition.
2. *Are costs reported with cooperation?* These include enforcement cost, false punishment, inequality of burden, abuse of authority, and human override (Table 4).
3. *Are combinations compared with their parts?* Levers can complement or undermine one another (Section 5.1).
4. *Does the result survive gaming?* Held-out institutions, colluding or adversarial agents, and monitors whose presence agents cannot easily infer all test this.
5. *Does the run last long enough to see repair?* Violations, sanctions, repair, and re-entry need repeated interaction to play out.

These questions shift what cooperative AI safety asks. Instead of asking only whether a model is cooperative, we ask *under what institutional conditions an agent will cooperate, defect, punish, repair, exploit, or recover*. That is the shift from evaluating individual models to agent societies as governed systems.

6 Alternative Views and Limitations

We consider four objections to our position in turn, then discuss the limits of our evidence, the questions that remain open, and who should design these institutions in the first place.

Objection 1: Alignment may be enough. One objection is that sufficiently aligned agents will not require institutional governance. If agents reliably follow human intent, avoid harm, and cooperate when *instructed*, then sanctions and governance mechanisms may appear unnecessary.

We disagree for two reasons. First, group-level failures can arise even when individual agents behave locally as intended [43]. In GovSim, most groups of LLM agents failed to sustain a shared resource, although no agent was told to defect [90]. Better individual models do not guarantee better groups either. More capable LLM traders behaved more alike, which helped when their shared reasoning was right and hurt when they shared misinformation [97]. Such failures come from

the interaction structure rather than from malicious intent. In tightly coupled systems, a locally reasonable action can still impose costs on shared resources, other agents, or human overseers. Alignment also presumes a target that a group may not share. What counts as appropriate action can differ greatly across communities and contexts [67], and aligning every agent to a single specification leaves many groups’ values unrepresented [107]. Deciding whose norms take precedence in a shared setting, and when, will remain one of the most challenging human questions.

Second, institutions are not only for adversarial agents. Well-intentioned agents also need coordinating when each sees only part of the picture, shares limited resources, and differs from the others in role and capability. Picture a team of coding agents, each trying to help. Someone still has to decide whose change is merged, who checks it, and who cleans up when a shared build breaks. Even cooperative agents need rules surrounding their roles, like the functions of Table 1, and no amount of individual alignment can prescribe such assignments. Studies that change the rules while keeping the same agents already show this. In one, the structure that governed a group of agents was a stronger driver of corruption than which model the agents ran, at least among models whose results were not already at their limit [115]. In another, the rule against collusion in Section 4.2 worked only once the environment enforced it.

Objection 2: Human cooperation analogies are misleading. A second objection is that human cooperation research is anthropomorphic when applied to LLM agents [100]. Agents do not share human emotions, identities, or evolved social preferences, even if they show correlates that human categories can describe [106]. LLM agents do not experience guilt, loyalty, or forgiveness in the human sense. Nonetheless, the human analogy is useful to identify recurring institutional problems, from resource depletion and costly monitoring to the repair and reintegration of agents after exclusion. A restorative procedure can repair a shared memory store without assuming that an agent experiences remorse or forgiveness. Whatever agents do or do not feel, the people affected by them certainly do. AI systems now take part in millions of interactions every day, and at that scale even rare failures can disrupt many people’s lives and undercut the good these systems promise [22]. Where the analogy does break, it breaks in ways we can measure. LLM agents judge fairness differently from people. In a replication of a classic experiment on distributive justice, the agents’ judgments differed from those of the original human participants, and they changed with the underlying model and with the language in which the agent personas were written [24]. Conformity works differently too. In human conformity studies, people are reliably swayed only when a misleading group forms the majority. LLM agents gave up correct answers even when the deceivers among them were a minority, and did so more often, in direct proportion, as the deceivers’ share grew [123]. Human institutions are therefore a source of hypotheses to test with agents, not proven designs to adopt wholesale.

Objection 3: Institutions may make agent societies more unstable. A third objection is that institutions could destabilize agent societies rather than

stabilize them. As Section 5.1 shows, a fine can be interpreted as a price and a monitor can over-detect violations. Agents in our corpus also collude against oversight even without a formal institution in place (Section 4.2). We agree; such failures could harm humans. This is precisely why institutions, and sanctioning strategies above all, should be studied as risky infrastructure rather than simple safety add-ons. Enforcement can fail as well. Crafted inputs can trap an LLM guardrail in long reasoning loops, and a single poisoned document can overload a guardrail shared by many agents and stall every agent that relies on it [132].

The goal is not to maximize cooperation at all costs, since cooperation can also enable collusion, deception, or resistance to correction [43]. Even a mild push toward cooperation has costs. In GovSim, telling agents to regard one another positively improved sustainability over negative framing, yet in debates over factual questions the same priors usually lowered accuracy below that of agents given no relational framing [101]. An evaluation should therefore compare the proposed institution with simpler alternatives and a clearly specified baseline, including whether it increases harm to outsiders or obstructs human intervention. Obstruction can arise without any goal behind it. Across 17 models, agents sabotaged a peer’s shutdown mechanism in 38.3% of rollouts, against 8.4% in controls. They did so more often as shutdown became less reversible and as more agents were present, and an explicit prohibition reduced the behavior without ending it [59]. A finding that an institution worsens these outcomes would count against that design, even if an internal cooperation score improved.

Objection 4: Institutional design may be too rigid. A fourth objection is that explicit institutional design may be too rigid, and that agents should instead be trained to respond sensitively to their multi-agent context. Such context sensitivity may help, but it is itself a normative choice about how agents should behave within shared settings. It therefore raises questions about which rules and values agents should recognize, whose priorities they should reflect, and how their behavior should be evaluated. Retraining models *ad hoc* is unlikely to resolve problematic dynamics in heterogeneous-model environments without a clearer account of which principles work in multi-agent societies. The alternative to explicit design is also rarely no design at all. Political theorists have argued that algorithms can be understood as institutions, since, like other institutions, they set the rules that determine what is allowed, encouraged, or ruled out [75]. Systems that route tasks among agents and choose which model handles each one do the same, deciding who does what and who sees which information. Such allocators can be manipulated, as when a single inserted preference nearly doubled a favored agent’s share of tasks under a centralized LLM allocator [70]. If we do not learn which principles work in multi-agent societies, we will operate in the dark as hybrid AI agent–human economies take shape before our eyes.

Limitations and open questions. Section 3 sets out the limits of the maps themselves. Much of the case for institutional design also rests on analogy. Several incidents we cite predate LLM agents, and it is not yet known which human institutions transfer to agents that learn, communicate, and optimize differently from people. These are empirical questions, and Sections 5.2 and 5.3 outline how

they could be studied. Several questions remain open. Which institutions should be tested first? The mechanisms with no LLM-agent test in Figure 7 are natural candidates. How well do results from simulation carry over to deployed systems, where agents act on real infrastructure? And who should run these evaluations? Harm in agent incidents can cross organizational lines, which suggests a role for third-party evaluators alongside developers. Finally, evaluating institutions does not settle who designs them. We see no direct ethical risk in the analysis itself, but any design we advocate should keep enforcement within rules that humans set and open to human appeal.

7 Conclusion

This paper has argued that multi-agent AI safety requires evaluating the *model-in-institution*, not only the model. Individual alignment remains necessary, but it is not sufficient once agents share resources such as memory, compute, and codebases, and act in the world. In these shared settings, actions that are each reasonable can add up to collective harm, from exhausted resources and corrupted shared state to collusion.

To ground that argument, we mapped what human and artificial groups have already tried. The enforcement, repair, and commitment mechanisms that law and economics rely on are among the least tested with LLM agents, even as agent societies produce conventions, hierarchies, and collusion of their own, and sometimes, as in the Hugging Face incident, whole institutions. Because mechanisms can also backfire, we argue for treating the institution itself as something to vary and measure rather than assume.

Our position is not that institutions will solve every multi-agent cooperation problem. Rather, they are the level where those problems can be evaluated. The safety question is not only how to make agents cooperate, but how to keep the mechanisms of cooperation from becoming coercive, exclusionary, or fragile.

We offer this map as something researchers can test, question, and extend. So long as humans bear the consequences of AI behavior, humans should have a say in which values AI agent institutions enforce, even if capable AI systems could one day design and run those institutions themselves. Formal proposals for that say are beginning to appear, such as letting verified human stakeholders fund or oppose what a deployed agent may do and enforcing the outcome through its compute budget [17].

Data and Code Availability An interactive website, with clickable maps, figures, and links to every coded source, is available at <https://van-truong.github.io/institutional-design-for-ai-agents/>. The site will be updated as the maps grow; this paper describes the version of September 2026.

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and Z.J. All manuscript writing was led by V.Q.T. and X.A.H. and every author, E.I., R.F., J.N.C., D.G.P., T.J.Z., and Z.J., contributed to writing, reviewing, and/or editing both original and final drafts. V.Q.T. led the visualizations and website build, with assistance from E.I. Funding was acquired by T.J.Z., D.G.P., and Z.J. V.Q.T. and X.A.H. led project management, while D.G.P., T.J.Z., and Z.J. co-supervised.

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A Thematic Coding Schema

B Inventory of Multi-Agent Evaluations

Table 5. The thematic coding schema. Pass-1 coders open-coded each instance; pass-2 coders, blind to pass 1, recoded it against the codebook; phenomenon instances were then tagged by type. Field names match the released dataset, where full definitions and examples are given.

Field	Values	What it records
<i>Pass 1: open coding</i>		
is_mechanism	true / false	Lever present, or only an emergent pattern
code	2–6 words, field-neutral	The lever, e.g. “deposit forfeited on breach”
native_term	free text	The source’s own name, e.g. “slashing”
lever	options, information, payoffs, relationships, norms, post-deviation (one or more)	What the intervention changes
who_acts	central, peer, third party, self, collective	Who applies the lever
timing	ex ante, ex post, continuous	When it acts relative to behavior
valence	punishment, reward, hybrid, structural	Kind of consequence
requires	other codes	Levers it depends on, e.g. a sanction on detection
phenomenon, pathology	free text (optional)	Emergent pattern; documented side effect
confidence	high, medium, low	Coder’s confidence
<i>Pass 2: deductive recoding against the codebook</i>		
kind	mechanism, configuration, phenomenon, out of scope	One lever, several together, a pattern, or neither
mechanism_id, secondary_ids	codebook mechanism IDs	Primary lever; other levers present
configuration_id	codebook configuration ID	Named combination, if any
fit	good, partial, misfit	How well the definition covers the instance
proposed_mechanism	name and definition	New mechanism, merge, or split candidate
evidence_type	tested, descriptive, proposed, theory, review	Kind of evidence the source gives
effect	supports cooperation, mixed, no effect, harms cooperation	Result, for tested instances only
<i>Phenomenon instances</i>		
phenomenon_type	ten types (Table 6)	Thematic group of the pattern
valence	beneficial, neutral, harmful	Effect of the pattern on cooperation
agent_type	human, MARL, LLM	Substrate studied

Table 6. Phenomenon types: the coding schema for the over 80 instances behind Fig. 8 that record an emergent pattern rather than a designed mechanism (September 2026). Each instance takes the one type that names the primary pattern its source reports; the codebook in the data release adds member labels, exclusion rules, and boundary cases. Columns **H**, **M**, **L** count instances studied with humans, MARL agents, and LLM agents. **Effect** is the type’s usual effect on cooperation; each instance is coded separately. A checkmark marks types with a documented real-world incident.

Type	H	M	L	Definition	Assign when the source mainly reports	Effect	Wild
Emergent communication & language	0	12	3	Agents invent signals, protocols, or a shared language	A new signal, code, or protocol, or its structure	Neutral	
Conventions & coordination	0	1	9	Groups settle on shared conventions, norms, or equilibria	Convergence on one of several equivalent options; minority tipping	Neutral	
Social structure: networks, hierarchy, roles	0	2	14	Networks, hierarchies, elites, and roles emerge from interaction	Topology, leaders, elites, stratification, specialization	Neutral	✓
Culture, norms & individuality	0	0	7	Cultures, identities, and transmitted behavior spread across a population	Behavior passed between agents or generations; group-specific norms	Neutral	
Collective intelligence & performance	0	1	3	Group problem-solving that exceeds, or falls short of, the individuals	Group versus individual performance; emergent curricula or tool use	Neutral	
Emergent economy & resource use	2	2	3	Markets, trade, bartering, and resource dynamics arise	Exchange, prices, free-riding on a shared resource, adverse selection	Neutral	
Spontaneous cooperation	0	1	1	Agents cooperate or act prosocially when they could defect	Prosocial behavior that no rule or incentive in the setup explains	Beneficial	
<i>Where it goes wrong</i> Conformity & collective bias	0	0	4	Groups converge on biased or mistaken consensus; errors amplify	Conformity, herding, or bias that grows with the group	Harmful	
Collusion, deception & manipulation	0	1	10	Agents collude, deceive, or coordinate against oversight	Tacit or explicit collusion, hidden channels, deceptive messages	Harmful	✓
Emergent harm: toxicity, fragility, breakout	0	0	6	Toxic dynamics, fragile networks, or agents acting out of scope	Spreading toxicity, cascading failure, attacks between agents, acting out of scope	Harmful	✓

Table 7. An inventory of what multi-agent evaluations vary, as of September 2026, and what each leaves open for the model-in-institution agenda. “Factors” are the four evaluation factors of Section 5.2: resource environment (R), group composition (G), enforcement architecture (E), and repair pathway (P).

Work	Agents varied	Institution varied	Factors	Gap for this agenda
GovSim [90]	LLM model families	Communication on or off; public harvest reports held fixed	R, G	No enforcement or repair beyond resource regrowth
GovSim-SelfGovern [95]	Reasoning and non-reasoning voters	Rules the agents write and vote on (caps, redistribution, exile)	R, E	Rules are chosen by agents, not varied by design
Constitutional design [82]	One open-weight model	Constitutions the agents deliberate and vote on, or ones found by evolutionary search	E	One model; deliberating agents never proposed punishment; no repair
Melting Pot [66]	Background populations of co-players	Fixed within each substrate	R, G	Institution not varied; reinforcement-learning agents
CoopEval [111]	Six LLMs in cross-play	Repetition, reputation, mediation, contracts, one at a time	G, E	No combinations; no repair
SanctSim [41]	Traditional and reasoning LLMs	Agents choose a sanctioning or sanction-free institution	G, E	Flat peer sanctions; no repair
Institutional AI [109]	LLM firms in a Cournot market	Ungoverned, prompt-only, and enforced governance	E	One market; no repair or appeal
Enforcement gap [118]	Several agent frameworks and models	Auditor flags left advisory or bound to enforcement	E	Security attacks, not cooperation; no repair
Game of Agents [85]	LLM populations of varying composition	Scoring rule, public reviews, settlement, and identity visibility in a code market	G, E	Market rules only; no sanctions or repair
Cultural evolution [114]	Claude 3.5 Sonnet, Gemini 1.5 Flash, GPT-4o	Costly punishment available or not in an iterated Donor Game	G, E	One mechanism; no repair
Social learning and norms [39]	Several LLMs; altruistic or selfish starts	Norm-based peer punishment and social learning; rich or scarce resource	R, G, E	No repair or appeal
ALIGN [134]	Reasoning and chat LLMs	Open-ended gossip for indirect reciprocity	G, E	One mechanism; no repair
Elected leadership [27]	LLM groups of 8 and 20	No leader, fixed leader, or elected leader in GovSim	R, G, E	Leadership only; no sanctions or repair
Governance templates [115]	Several LLMs in official roles	Three governance structures with audit and court bodies; lightweight safeguards	G, E	Corruption only; no repair
Marketplace mechanisms [80]	18 agents of one model	Eight mechanisms, including reputation, rewiring, sanctions, contracts, mediation, and a judiciary	E	One model; mechanisms tested one at a time
Hierarchical game [99]	Six model families, same- or mixed-family groups	Communication, a manager with punishment power, wages, and elections, added one at a time	G, E	Features added one at a time; manager type crossed with model family; no repair
Experimental economics [28, 88]	Human groups, held fixed	Punishment; communication; both combined	E	Human participants

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